

Symmetry in space

Central, axial and plane symmetry — the three kinds, their coordinate rules, and how to count them on a solid.

LESSON 10–11

2 × 40 MIN

GEOMETRY 11, §6

IGCSE E4.6

LESSON CLOCK

10'

20'

24'

20'

14'

2

Three kinds	10 min
Coordinate rules	20 min
Counting on solids	24 min
Symmetry as a shortcut	20 min
Practice	14 min
Homework	2 min

LEARNING OBJECTIVES

- Distinguish central, axial and plane symmetry.
- Write the coordinate rule for each and apply it.
- Count the symmetries of a given solid.
- Use symmetry to shorten a calculation or a proof.

TEXTBOOK

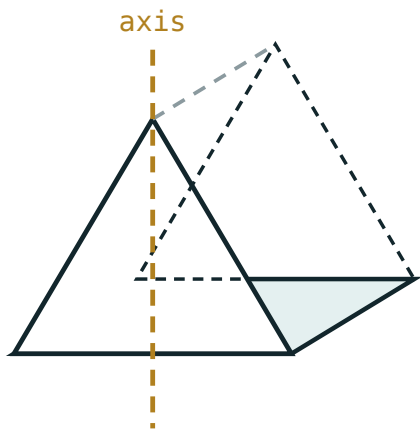
National	pp. 53–64
Cambridge	Core & Extended, pp. 355–362
Workbook	Exercise 6.1

01

Explanation

Three kinds of symmetry

KIND	FIXED OBJECT	RULE (ABOUT THE ORIGIN / AXES)
central	a point	$(x, y, z) \rightarrow (-x, -y, -z)$
axial (order 2)	a line	$(x, y, z) \rightarrow (-x, -y, z)$ <i>about Oz</i>
plane (mirror)	a plane	$(x, y, z) \rightarrow (x, y, -z)$ <i>in $z = 0$</i>



a plane of symmetry cuts it into mirror halves

*The same cuboid, seen through a plane of symmetry
and turned about an axis.*

CENTRAL SYMMETRY IS A HALF-TURN PLUS A MIRROR

In space, $(x,y,z) \rightarrow (-x,-y,-z)$ equals a 180° rotation about Oz followed by a reflection in $z = 0$. It reverses orientation, so it is not a rotation.

Counting on solids

SOLID	CENTRE?	PLANES	AXES
cube	yes	9	13
cuboid, edges different	yes	3	3
regular tetrahedron	no	6	7
square-based pyramid	no	4	1
regular hexagonal prism	yes	7	7
sphere	yes	infinitely many	infinitely many

A PYRAMID HAS NO CENTRE OF SYMMETRY

The apex would have to map to a point of the base, and there is nothing there. Any solid with a centre has every point paired with an opposite one — so a pyramid, a cone and a tetrahedron all fail.

Symmetry as a tool

Three uses that save real work:

- The **centre of mass** lies on every plane and axis of symmetry — so at their intersection.
- Two lengths or angles symmetric about a plane are **equal**, with no calculation.
- A section perpendicular to an axis of order n repeats n times, so one computation covers n cases.

Example. The centre of a cube with opposite vertices $(0,0,0)$ and $(6,6,6)$ is $(3,3,3)$ — by symmetry, in one line, with no distance formula.

02 Worked examples

EXAMPLE 1 Find the image of $(2, -3, 4)$ under central symmetry about the origin, and about $(1, 1, 1)$.

① About O : negate all.
 $(-2, 3, -4)$

② About C : $P' = 2C - P$.

③ $(2 - 2, 2 + 3, 2 - 4)$

ANSWER

$(-2, 3, -4)$ and $(0, 5, -2)$

EXAMPLE 2 How many planes of symmetry has a regular hexagonal prism?

① The hexagon has 6 lines of symmetry.
 6 vertical planes.

② One horizontal plane halfway up.

ANSWER

7

EXAMPLE 3 Does a square-based pyramid have a centre of symmetry?

- ① A centre pairs every point with an opposite.
- ② The apex would need a matching point below the base.
There is none.

ANSWER

No

03 **Interactive model**

Hold a cuboid box and find each plane, then each axis, and count aloud.

Counting symmetry in space

Central symmetry sends $(1,2,3)$ to:

$(-1,2,3)$

$(1,-2,3)$

$(-1,-2,-3)$

$(3,2,1)$

A cube has how many planes of symmetry?

3

6

9

12

A square pyramid has a centre of symmetry:

yes

no

sometimes

only if regular

A regular hexagonal prism has how many planes?

6

7

12

13

The centre of mass lies on:

no axis

every plane of symmetry

the base

the apex

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Central symmetry	Markaziy simmetriya	Центральная симметрия
Axial symmetry	O'qqa nisbatan simmetriya	Осевая симметрия
Plane symmetry	Tekislikka nisbatan simmetriya	Зеркальная симметрия
Centre of symmetry	Simmetriya markazi	Центр симметрии
Axis of symmetry	Simmetriya o'qi	Ось симметрии
Plane of symmetry	Simmetriya tekisligi	Плоскость симметрии
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Symmetric figure	Simmetrik shakl	Симметричная фигура
Symmetry group	Simmetriya guruhi	Группа симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Central symmetry about C sends P to:

Reflection in $z = 0$ changes:

Which solid has no centre of symmetry?

A cuboid with all edges different has how many axes?

0

1

3

9

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Central image of $(1,2,3)$

ANSWER $(-1,-2,-3)$

2. Plane image of $(1,2,3)$ in $z = 0$

ANSWER $(1,2,-3)$

3. Axial image about Oz of $(1,2,3)$

ANSWER $(-1,-2,3)$

4. Planes of symmetry of a cube

ANSWER 9

5. Planes of a $2 \times 3 \times 4$ cuboid

ANSWER 3

6. Has a sphere a centre of symmetry?

ANSWER yes

7. Has a cone a centre of symmetry?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. Central image of $(2, -3, 4)$ about $(1, 1, 1)$

ANSWER $(0, 5, -2)$

2. Planes of a regular hexagonal prism

ANSWER 7

3. Planes of a regular tetrahedron

ANSWER 6

4. Axes of a cube

ANSWER 13

5. Planes of a $3 \times 3 \times 8$ cuboid

ANSWER 5

6. Centre of a cube with opposite vertices $(0, 0, 0)$, $(6, 6, 6)$

ANSWER $(3, 3, 3)$

7. Rotational order of a regular pentagonal pyramid

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. Prove central symmetry equals a half-turn followed by a mirror

ANSWER $(x,y,z) \rightarrow (-x,-y,z) \rightarrow (-x,-y,-z)$

2. Why has a regular tetrahedron 6 planes but no centre?

ANSWER Each plane holds an edge and bisects the opposite; no vertex has an opposite

3. Find the centre of symmetry of the cuboid with opposite corners $(1,2,3)$ and $(7,10,15)$

ANSWER $(4,6,9)$

4. A solid has exactly one plane of symmetry. Sketch one

ANSWER e.g. an isosceles-triangle prism

5. How many symmetries (rotations only) has a cube?

ANSWER 24

6. Show that a solid with a centre of symmetry has an even number of vertices

ANSWER They pair up

7. Use symmetry to find the centroid of a regular tetrahedron of side a

ANSWER At $\frac{1}{4}$ of the height above the base centre

07 Homework

Homework — 4 tasks

Sketch each solid and mark the symmetries on the sketch.

- Find the image of $(3, -2, 5)$ under central symmetry about $(0,0,0)$ and about $(2, 1, 3)$.
- Count the planes and axes of symmetry of a cube, a regular tetrahedron and a $4 \times 4 \times 9$ cuboid.
- Explain why a pyramid can never have a centre of symmetry.
- Use symmetry to find the centre of the cuboid with opposite corners $(-1, 0, 2)$ and $(5, 8, 10)$.